From a Wall-Following Car to a Data-Producing Robot: Rebuilding a 1/10-Scale Platform as Substrate for a Learned Driving Policy

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Introduction

This project picks up the 1/10-scale self-driving car in the SysLab annex where the 2024 Senior Research work left it [9]. That report ended with a car that could follow a hallway wall using a hobby ESC, a PWM breakout board and the DonkeyCar library, and it recommended that the next group not deviate from the F1TENTH bill of materials, because every substitution costs weeks.

What I want to build is a vision-language driving policy: take an open vision-language-action or vision-language-navigation checkpoint (Alibaba's Qwen-Robot suite [17] is the current example, though the plan works with any of them) and post-train it on data from this car, so that it follows natural-language goals like "go to the end of the hall and stop before the door" instead of a precomputed raceline. A second car, built from the same parts list, then opens up agent-to-agent experiments: two vehicles sharing pose and intent over the network.

That is not where most of this year went, and the gap is what this report is about. A policy like that is only as good as the robot underneath it, and the robot I inherited could not have produced usable training data. It could not report how fast its wheels were turning, so an "action" recorded during a demonstration would have been an uncalibrated PWM pulse. It had no inertial measurement, so there was no proprioceptive state to condition on, and no way to tell a skid from a straight line. It could only be driven by a person standing next to it with a wired gamepad, which caps a dataset at the length of a USB cable and one operator's patience. And it was a single physical artifact whose configuration lived in one laptop, so a second car meant reverse-engineering the first.

The work in this report is therefore about the platform, and I have organized it that way: each change is justified by what it makes possible later, not by the lap time it saves.

1. **Actuation with feedback.** Rebuild the drivetrain around the VESC motor controller the previous report gave up on, so commands are currents and speeds with measured consequences. That is the label side of a demonstration.

2. **Sensing in the policy's coordinate frame.** Bring the lidar and camera into ROS 2 in the shape the stack expects, including the camera's inertial unit, which supplies proprioception, de-skews the lidar, and feeds the traction governor.

3. **Teleoperation as a data channel.** Make the car drivable from a browser over its own network, so demonstrations can be collected by anyone, anywhere, with the same message the policy will later emit.

4. **Reproducibility as infrastructure.** Put the configuration, firmware state and scripts in the repository, so the second car is a clone instead of a rebuild.

Perception is where the two halves meet. I trained and deployed a car detector on the platform, which is useful for head-to-head racing now and also a rehearsal of the train-offline, deploy-on-the-robot loop the pivot needs. All of it is checked into the team repository [2]. What I have *not* done is validate the autonomous controller on the floor with the new drivetrain. The competition that was the deadline (Boston, September 6–7, 2026) was cancelled when most of the team withdrew, which is also what freed the project to pivot. The Results section separates what was measured from what still needs a track, and the Pivot section states what remains before a policy can be trained.

Prior Approaches

Three earlier designs shaped the choices here.

The 2024 report [9] drove the same chassis through a PCA9685 PWM generator into an 80 A hobby ESC. That path works with DonkeyCar and needs no motor controller protocol, but it is open loop: the ESC reports nothing back, so there is no wheel speed, no current, no temperature, and no way to calibrate throttle except by driving. The report describes spending several months trying to control a VESC over USB and through an Arduino before switching to the hobby ESC.

The F1TENTH reference stack [7] uses the VESC over USB with a ROS 2 driver, converts Ackermann commands into VESC RPM set-points, and reconstructs odometry from the VESC's electrical RPM. It expects a Hokuyo lidar on Ethernet and a specific gamepad. The Jetson in the lab already had this stack installed, but with a lidar launch file for a sensor the car does not have and a control mode that had caused a motor to overheat.

The team's own AtlasAutoware stack [3] contributes the planning and control layer: a linear time-varying model predictive controller for raceline tracking with a persistent warm-started quadratic program (0.6 ms mean solve time in the reported benchmark), a Model- and Acceleration-based Pursuit fallback, minimum-curvature raceline refinement, a friction-limited velocity profile, an IMU-based traction governor, actuation-latency compensation and a Frenet-frame overtaking planner. Those results were obtained in the simulator against a kinematic bicycle plant. The hardware layer in that paper assumed an OAK-D Pro camera; ours is an Orbbec.

MuSHR [1] is the other common open racecar design and uses the same VESC interface, which is one reason to adopt it: any VESC-based stack becomes usable on this car.

System Architecture

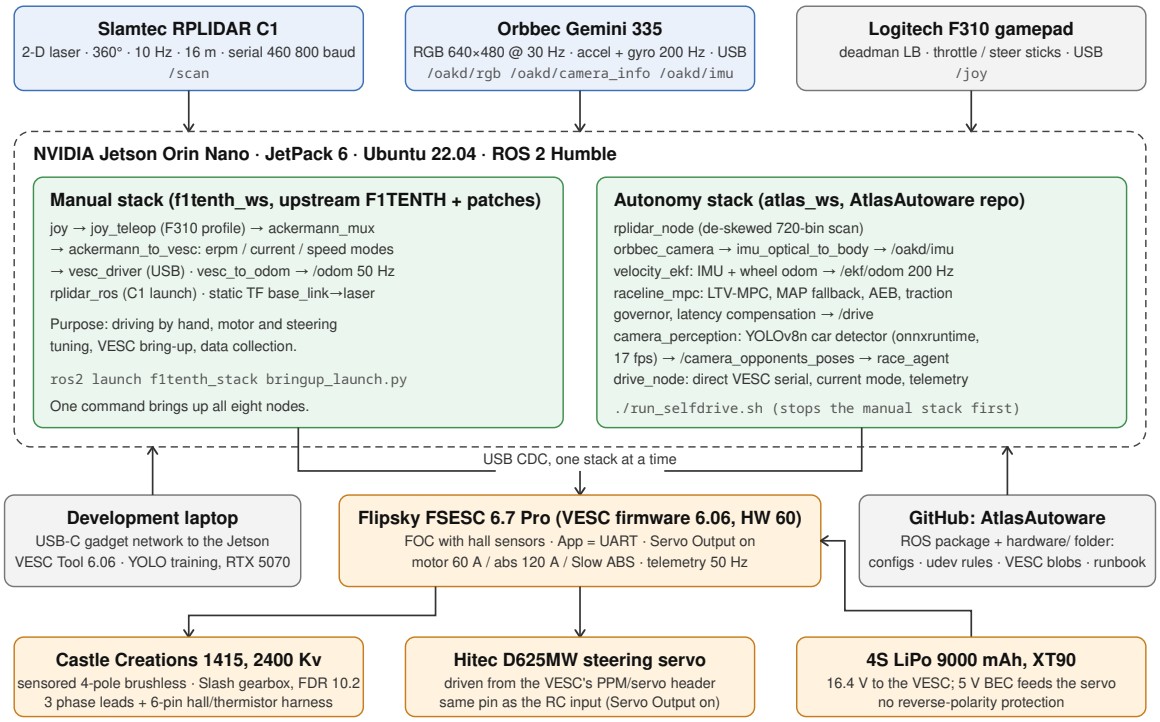


Figure 1: The car as built. Blue: sensors. Green: the two ROS 2 workspaces on the Jetson. Amber: the power and actuation chain. Grey: the development laptop and the repository. Only one workspace talks to the VESC at a time; the autonomy launcher stops the manual stack before it starts. Read as substrate for the pivot: the sensor row is the observation space, the amber chain is the action space with feedback, and the manual stack's teleoperation path is where a policy server later attaches (Figure 3).

Figure 1 shows the system. Two ROS 2 workspaces live on the Jetson and share the same hardware. The manual stack is the upstream F1TENTH system with our patches: a gamepad drives the car through the F1TENTH Ackermann mux and VESC driver, and it is what I use to tune the motor controller and to drive the car by hand. The autonomy stack is the AtlasAutoware package. It has its own lidar node (which de-skews each scan using the velocity estimate), its own VESC driver, the velocity EKF, the MPC racing node and the camera perception node. The two stacks cannot both own the VESC's USB port, so the autonomy launcher kills the manual stack before starting.

Table 1 lists the hardware. Compared with the 2024 build, the ESC, motor, camera, lidar and battery are all different; the chassis, Jetson and servo are the same.

Part	Notes
NVIDIA Jetson Orin Nano	JetPack 6 (L4T R36.5), Ubuntu 22.04, ROS 2 Humble. Reached from the laptop over the USB-C gadget network at 192.168.55.1.
Flipsky FSESC 6.7 Pro	VESC firmware 6.06 on the HW 60 profile. Replaced once after reversed battery leads destroyed the first board (the VESC has no reverse-polarity protection).
Castle Creations 1415, 2400 Kv	Sensored 4-pole brushless motor, 5 mm shaft, with the hall/thermistor harness wired to the VESC's 6-pin sensor port. Replaced the stock Traxxas Velineon 3500.
Traxxas Slash drivetrain	Final drive ratio 10.2, 0.109 m tires, on a laser-cut deck.
Hitec D625MW	Steering servo on the VESC's servo header.
Slamtec RPLIDAR C1	2-D lidar, 360°, 10 Hz, 16 m, 460 800 baud over a CP210x serial adapter [18].
Orbbec Gemini 335	RGB-D camera with a 200 Hz accelerometer and gyroscope [13]; used here for colour and the IMU, depth off.
Logitech F310	Wired gamepad, XInput mode.
4S LiPo, 9000 mAh, XT90	16.4 V measured. The VESC's 5 V BEC powers the servo.

Table 1: Hardware on the car as of September 2026.

Stage 1: Actuation with feedback

A demonstration is a pair: what the world looked like, and what the driver did. The second half is worthless if "what the driver did" cannot be measured. The 2024 car commanded a hobby ESC with a unitless PWM pulse and got nothing back, so its own report notes that throttle could not be calibrated and that the same pulse produced different speeds on different surfaces. A policy trained on those labels would be learning a mapping that does not exist. The VESC gives the action space real units: currents and speeds with measured consequences.

Getting the VESC to drive the motor

The VESC is attractive because it closes the loop: it measures phase currents, estimates rotor speed, exposes a PID speed controller, and reports voltage, current and temperature over USB at 50 Hz. It is also unforgiving. Three problems consumed most of this stage.

The first was motor control itself. With the stock sensorless Velineon motor, the VESC's field-oriented control depends on a back-EMF observer that has nothing to observe at standstill. Below roughly 2000 electrical RPM the motor cogged, drew current without moving, and twice heated its phase wires until solder joints failed. Tuning helped (the working configuration used an observer gain of 62, a sensorless threshold of 2000 electrical RPM, motor current limits of 50–60 A and the firmware's slow absolute-current limit, without which current ripple tripped the over-current fault), but the underlying dead zone was physical. I replaced the motor with a sensored Castle 1415 whose hall sensors give the controller rotor position from zero speed. Detection with hall sensors produced a hall table and motor parameters that the firmware's observer tracked cleanly up to the 10 000 electrical RPM I tested on a stand.

The second was the software's response to the same dead zone. Both stacks originally commanded the VESC in RPM mode. Someone on the team had already added a custom `erpm` control mode to the F1TENTH `ackermann_to_vesc` node that never commands below 3000 electrical RPM, and a matching `current` mode in the AtlasAutoware `drive_node` that maps the planner's speed to motor current with a feed-forward minimum of 10 A, so the car breaks stiction the instant a non-zero speed is requested instead of waiting for the RPM loop to wind up. With a sensored motor these are no longer necessary for safety, but current mode remains the default for the autonomy path because it gives the MPC direct torque authority. The speed-to-RPM gain that odometry depends on was also wrong in both stacks (it was the reference car's value, 4614); for this drivetrain it is $60 \cdot 10.2 \cdot 2/(\pi \cdot 0.109) \approx 3575$ electrical RPM per m/s.

The third problem is the one worth recording for whoever comes next, because it cost a day and had no error message. After the replacement VESC was installed and configured, the motor ran from VESC Tool on the laptop but not from ROS: every command from the Jetson produced zero duty, zero current and fault code zero. Rather than keep guessing from the ROS side, I wrote a small Python implementation of the VESC serial protocol (CRC-16/XMODEM packet framing, `COMM_GET_VALUES`, `COMM_SET_RPM`, `COMM_GET_MCCONF`, `COMM_GET_APPCONF`) and a decoder for the firmware's serialized configuration, taking the field order and the float16 scale factors from `confgenerator.c` in the 6.06 firmware source [19]. That made the cause visible in one read. A fresh Flipsky board ships with the RC-input application enabled (`app_to_use = PPM_UART`) and servo output disabled. On this hardware the PPM input and the servo output share one pin, so the steering servo's signal wire was being decoded as an RC receiver, and the PPM application overwrote every USB command with "neutral" fifty times a second. Setting `servo_out_enable`, switching the application to UART and rebooting the controller fixed both the motor and the steering at once. The same tooling then wrote the current limits (60 A motor, 120 A absolute, slow absolute-current limit on) and saved the complete motor and application configurations as binary blobs in the repository, so the next dead board can be restored in seconds without VESC Tool.

What the actuation layer now provides

From the manual stack, the F310 gamepad drives the car with a dead-man button. From the autonomy stack, `drive_node` auto-detects the VESC on its stable udev path, commands current or speed, publishes wheel-speed odometry at 20 Hz and logs battery voltage, MOSFET temperature and fault codes. On the bench the VESC held commanded speeds of 1500, 3500, 6000 and 9000 electrical RPM to within 1% at 1.5–3 A.

Stage 2: Sensing, observations and proprioception

A vision-language policy conditions on what the robot sees and, if it is to drive rather than sightsee, on how the robot is moving. This stage supplies both, in the coordinate conventions the rest of the stack (and any policy trained on its logs) assumes.

Lidar

The lidar is an RPLIDAR C1, not the A2 the launch files assumed. The C1 uses 460 800 baud, and the A2, A3 and S1 launch files all time out on it silently. Both stacks were

changed to the C1 settings; the F1TENTH bring-up now starts the lidar itself instead of requiring a second launch. The scan arrives at 10 Hz with 16 m range.

Camera and IMU

The AtlasAutoware stack was written for an OAK-D Pro and consumes three topics from it: a colour image, its calibration, and a single `sensor_msgs/Imu` message carrying both acceleration and angular rate. The important consumer is not vision. The `velocity_ekf` node fuses the IMU with wheel odometry into a 200 Hz velocity estimate that the lidar node uses to de-skew each 100 ms sweep, and the MPC's traction governor reads the gyro yaw rate to back off speed when the car slides. Without an IMU both features are inert.

I replaced the OAK-D node with the official Orbbec ROS 2 driver [14] (OrbbecSDK v2 branch, built on the Jetson) and remapped its colour topics onto the names the stack expects. The IMU needed more than a rename. The Orbbec reports acceleration and angular rate in the camera's optical frame (x right, y down, z forward), while the EKF and the governor assume ROS body axes (x forward, z up) [8]. Fed directly, the EKF would have integrated lateral acceleration as forward acceleration and treated roll rate as yaw rate. A 60-line relay node, `imu_optical_to_body`, rotates both vectors and their covariances. It also publishes with reliable QoS, because the EKF subscribes with the ROS 2 default and a best-effort publisher never matches it: the first version of the relay produced a 200 Hz stream that nobody received, with no warning beyond one line in the log.

Building the driver exposed a JetPack quirk that will bite any C++ package using OpenCV: the image had NVIDIA's OpenCV 4.8 development headers installed alongside Ubuntu's 4.5.4 libraries, so every `find_package(OpenCV)` failed. Downgrading the development package to 4.5.4 to match the runtime fixed it. The camera negotiates USB 2.0 with the cable currently on the car, which is enough for 640×480 colour at 30 Hz and the IMU at 200 Hz; depth would need the USB 3 cable.

With the car level and still, the rotated IMU reports gravity on body $+z$ at 9.75 m/s^2, and `/ekf/odom` publishes at 200 Hz whenever the VESC odometry is present.

Stage 3: Teleoperation as a data channel

The 2024 car was driven by a person holding a wired gamepad next to it. That is a hard ceiling on demonstration data: one operator, one room, one cable. It also means the human's commands and the eventual policy's commands would enter the system through different doors, so a policy could not be dropped into the same slot the human occupied.

I replaced it with a single ROS 2 node, `web_pilot`, and made the car its own access point. The node serves one page: the camera stream as MJPEG, keyboard (W/S throttle, A/D steering) and browser-gamepad input, a lidar plot, and link and battery telemetry. Commands are posted as JSON at 30 Hz and republished as `sensor_msgs/Joy`, so the unmodified F1TENTH teleoperation chain drives the car; a policy server emits exactly the same message. Only the tab that is actually driving sends commands, so other viewers cannot interfere, and if nothing arrives for 250 ms the node publishes neutral, which releases the dead-man and stops the car; the VESC's own one-second timeout sits behind that.

The network is selectable, because collecting varied data means driving in varied places. The car can be its own 5 GHz hotspot (highest bandwidth, tens of metres), its

own 2.4 GHz hotspot (roughly double the range), or a client on an existing mesh network, in which case it roams between nodes and the usable area becomes the mesh's footprint. Given an internet uplink (a tethered phone or an LTE modem) and a WireGuard overlay, the same page works from anywhere, at 60–150 ms of latency and a reduced video setting. A steering trim (the car pulls right, and mechanical adjustment did not remove it) is applied on the car to every input source and kept across restarts.

Two robustness details came out of real use. The VESC's USB port re-enumerates when the car is knocked hard enough, and the F1TENTH VESC driver aborts when its serial port disappears, which left the web page up and the car deaf; the driver, the Ackermann converter and the joystick node now respawn automatically. And an early version of the gamepad bridge published its axes with the SDL joystick sign convention instead of the ROS one, so throttle and steering came out mirrored. That is the same kind of frame mismatch as the camera IMU below, and a reminder that a data pipeline silently learns whatever convention error you leave in it.

Stage 4: Perception, and a rehearsal of the training loop

Why a car detector

The stack's head-to-head agent, `race_agent`, fuses opponent positions from the lidar with any that arrive on `/camera_opponents_poses`. The camera perception node that publishes that topic was already written: it runs a single-class YOLOv8 detector, turns each box into a range from the box width and the camera's focal length, and hands the result to the agent's opponent tracker. What was missing was a model; the repository contained an empty dataset scaffold and a training script that had never been run.

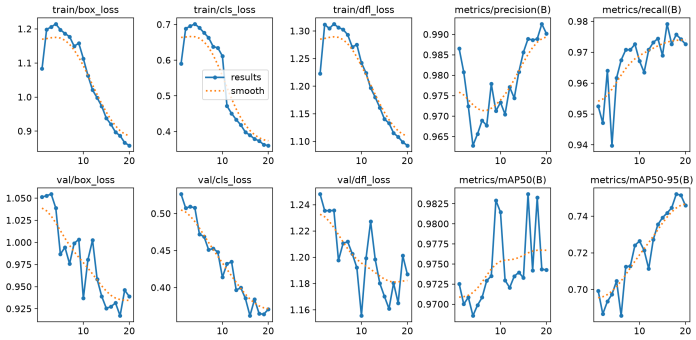
Data

No labelled images of F1TENTH cars existed in the lab, and there was only one car to photograph. I assembled a dataset from four public Roboflow Universe collections [6, 5, 15, 4], all CC BY 4.0, with a merge script that maps each source's class names (*car*, *Cars*, *f1tenth*, *racecar*) to one class, drops boxes for anything else, and keeps each source's own train/validation split so that the augmented copies in one collection never leak across the split. The result was 4907 training and 547 validation images with 5389 boxes.

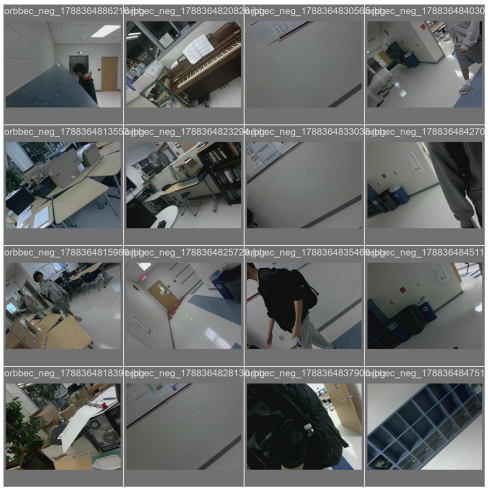
The first model trained on that data reached a mean average precision at IoU 0.5 of 0.974 on the held-out images, but it also called a hand or a face at the lens a car at close range in 21 of 185 test frames. None of the public images contain a person 30 cm from the camera. I therefore recorded 185 frames from the car's own Orbbec while walking it around the hallway and waving hands, tools and a backpack in front of it, added them to the training set with empty label files, and fine-tuned for 20 epochs. False detections on those frames dropped to 0 of 185, and the validation metrics did not suffer (Table 2). These negatives stay off GitHub because they contain people; the merge and training scripts in `tools/` rebuild everything else from the public downloads.

Model	mAP_{50}	$mAP_{50:95}$	Prec.	Rec.	False cars / 185 neg.
YOLOv8n, public data, 50 epochs	0.974	0.754	0.993	0.971	21
+ Orbbec negatives, 20-epoch fine-tune	0.983	0.752	0.989	0.976	0

Table 2: Detector results on the 547-image validation split (578 after adding a share of the negatives). Training on an RTX 5070 laptop GPU took 17 minutes for the first run and 9 for the fine-tune.



(a) Training curves for the fine-tune.



(b) Predictions on validation images.

Figure 2: Detector training. Validation images are from the public collections.

Running it on the Jetson

The perception node offered three inference backends: a TensorRT engine, OpenCV's CUDA DNN, and OpenCV on the CPU. None of them could run on this Jetson. The image has no CUDA toolkit or TensorRT installed, Ubuntu's OpenCV has no CUDA support, and OpenCV 4.5.4's DNN module fails on the YOLOv8 head's reshape layer. I added a fourth backend on ONNX Runtime [11], which installs from a wheel and needs nothing from NVIDIA. Through that path the 416-pixel model runs at 58 ms per frame (17 Hz) and the 640-pixel model at 124 ms on the Orin's CPU with four threads. On 40 real validation images the deployed 416 model found the car in every one.

Two guards were added in the node because a detector that is right 99% of the time is still dangerous when the 1% is a hand at the lens reported as an opponent at 0.2 m. Detections implying a range under 0.8 m or a box wider than 60% of the frame are dropped; inside that range the lidar owns the problem anyway. The confidence threshold was raised from 0.35 to 0.5. The node is off by default in the car launch and enabled with `use_perception:=true`, since it costs a CPU core and only the head-to-head agent uses it.

What the Architecture Buys

Most of the individual pieces exist elsewhere. What matters here is that each one removes a specific obstacle between this car and a learned policy.

Actions became measurable quantities. The 2024 build could only send a PWM pulse and hear nothing back. This car reports wheel speed at 20 Hz, battery voltage and temperatures, and accepts current commands directly. A logged demonstration is now a triple of observation, action and outcome, all in physical units, and the same closed loop is what let the current-mode launch fix the low-speed stall.

The camera is a motion sensor first. The Orbbec's most useful output is its 200 Hz inertial stream. It supplies the proprioceptive state a policy conditions on, it de-skews the lidar so scans stay accurate at speed, and it gives the traction governor a yaw rate. Vision is its second job.

One command channel for humans and policies. Teleoperation, the joystick, the Python client and any future policy server all publish the same `sensor_msgs/Joy`, behind the same watchdog. Swapping a human for a model changes the sender, not the rest of the system, and the safety envelope does not care which one is talking.

The dataset is not room-shaped. Browser teleoperation over selectable network modes means demonstrations can be collected by anyone with a link, in any space the network reaches, on any device. That is the difference between a few hundred episodes and a few thousand.

Firmware-level diagnosis, and firmware in the repository. Reading and writing the VESC's full configuration from the Jetson turned a silent failure into a one-line diagnosis, and it makes the motor controller's state a versioned artifact instead of something living in one laptop's settings.

Negatives from the deployed camera. Public datasets tell a detector what a car looks like. Only the target camera can tell it what the false alarms look like. Two minutes of recording eliminated all 21 false detections, which is the small version of a larger point: on-robot data beats more internet data.

Reproducibility as a deliverable. The `hardware/` folder holds the F1TENTH configuration overrides, the patch to the VESC driver, the udev rules that give the gamepad, lidar, VESC and camera stable names and permissions, the VESC configuration blobs and the tools that restore them, the network and launch scripts, and a runbook of the failures above. This is what turns "build a second car" into a parts order, and a second car is what makes any agent-to-agent work possible.

The Pivot: A Post-Trained Driving Policy

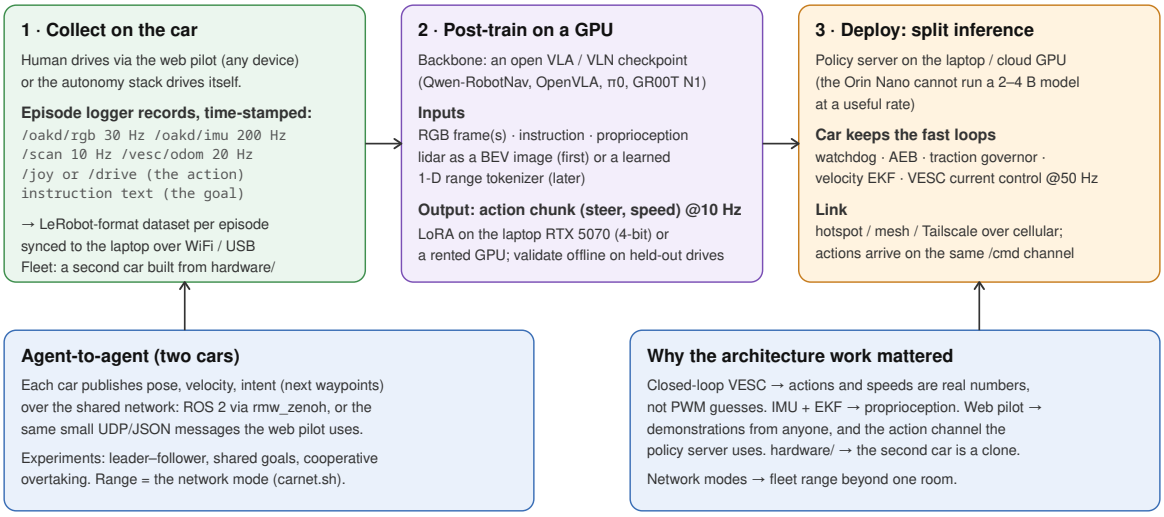


Figure 3: The intended pipeline. The platform work above supplies the left box (data with real actions and proprioception, collected by anyone over the network) and the safety and control loops in the right box that a remote policy cannot be trusted to provide.

What the policy would be

Recent robotics foundation models pair a vision-language backbone with an action head and are meant to be post-trained on a target platform's data. Alibaba's Qwen-Robot suite [17] is the current instance, releasing a navigation model, a manipulation model and a video world model together, and open vision-language-action checkpoints such as OpenVLA [10], π_0 [16] and NVIDIA's GR00T N1 [12] fill the same slot. The plan does not depend on the choice; it depends on the interface. All of them consume images, a natural-language instruction and some proprioceptive state, and emit a short chunk of future actions.

For this car the mapping is direct: the Orbbec's colour frames and the instruction text are the inputs, the velocity EKF's speed and yaw rate are the proprioception, and the action is the (steering angle, speed) pair the Ackermann chain already accepts. The lidar is the one input these models were not designed for. The cheap first approach is to render each scan as a small bird's-eye image and let the vision encoder see it, and the more principled second version is a learned range tokenizer. The training objective is imitation on teleoperated episodes, with the raceline MPC available as a second source of demonstrations that are smoother than a human's.

What has to be built

Three things, none of which need new hardware. First, an *episode logger*: a node that records synchronized camera, lidar, IMU, odometry and command streams into a standard imitation-learning format (LeRobot's dataset layout is the obvious target) with an instruction string per episode, plus a way to mark an episode good or bad without stopping. Second, a *policy server*. The 2–4 B-parameter models in this class will not run at a useful rate on an Orin Nano, so inference belongs on the laptop or a rented GPU, and

the car keeps every fast loop (watchdog, emergency braking, traction governance, the 50 Hz current loop) and treats the policy as one more source of `/cmd` messages under the same timeout. Third, *evaluation*: success rate on held-out instructions in the hallway, and offline action-prediction error on held-out drives so models can be compared without risking the car.

Two cars

The second car is what the `hardware/` folder was written for. With two vehicles on one network, each publishing pose, velocity and intended waypoints (over ROS 2 with a Zenoh middleware, or the same small JSON messages the web pilot already uses), the experiments become leader–follower, shared-goal navigation, and cooperative overtaking. The useful questions there are how much bandwidth an "intent" message actually needs and how gracefully behaviour degrades as the link gets worse. How far apart the two cars can be while that holds is set by the network mode from Stage 3, which is why that work sits in the report rather than in a footnote about convenience.

Where this can go wrong

The Orin Nano cannot host these models, so split inference is forced, and it puts a network link inside the control loop. That is fine at hallway speeds with a watchdog and not fine at racing speeds. Post-training a model this size on a laptop GPU means parameter-efficient fine-tuning and a small dataset, so a few hundred episodes from one hallway will produce a policy that is good at that hallway and not obviously anywhere else. And every VLA result I am building on is reported by its authors on their own platforms, mostly manipulators. Whether any of it transfers to a fast Ackermann vehicle with a 20 cm wheelbase is the open question, and the reason it is worth trying.

Results

Everything in Table 3 was measured on the car on 2 September 2026 with the wheels off the ground or the car stationary.

Quantity	Measured
Lidar scan rate (`/scan`)	10.0 Hz, 720 bins, 16 m
Camera colour (`/oakd/rgb`)	30 Hz at 640×480; 18–28 Hz with the detector running
Camera IMU (`/oakd/imu`)	195–200 Hz; gravity on body $+z = 9.75\,\mathrm{m/s^2}$
Velocity EKF (`/ekf/odom`)	200 Hz
VESC telemetry / wheel odometry	50 Hz / 20 Hz; 16.4 V, MOSFETs 26–29 °C, fault 0
Speed tracking on a stand	commanded 1500 / 3500 / 6000 / 9000 eRPM; measured 1472 / 3493 / 5986 / 9006
Gamepad path, manual stack	full throttle held 10 000 eRPM at 3–4 A; steering full range
Detector on the Jetson CPU	58 ms per frame at 416 px; 0 false detections in 55 s of hands and faces
Web teleoperation	command round trip 1–2 ms over the USB link; commands republished at 50 Hz; steering follows to the servo limits; neutral within 250 ms of the last packet

Table 3: Hardware measurements.

What is not in that table matters as much. The autonomous controller has not driven a lap on this drivetrain. The steering servo's centre and end-point values in the configuration are still the reference car's; the wheelbase and the lidar and camera mounting offsets in the static transforms are estimates, not measurements; the speed-to-RPM gain has been computed but not checked by driving a known distance; and the detector has only ever seen other F1TENTH cars in photographs, so its range estimate, which assumes a 0.30 m car width, is unverified against a real opponent. On the teleoperation side, only the car's own 5 GHz hotspot and the USB link have been driven through; the mesh-client, tethered and overlay-network paths are implemented and documented but not yet exercised, and no latency figures over cellular are claimed here. Nothing about the learned policy has been built; the Pivot section is a plan, not a result.

Next Steps

The immediate platform work is unchanged and short: drive a measured five metres and compare the odometry before trusting any autonomy; calibrate the servo the way the 2024 report did (a range finder on the front wheel) and enter the centre and limits in both stacks' configuration, which also removes the right-hand pull that the web trim currently masks; measure the wheelbase and the two sensor offsets. Install the JetPack CUDA and TensorRT packages when the Jetson is on a network that is not a captive portal; the perception node already prefers a TensorRT engine if one is present, and the 640-pixel export is in the repository for that purpose.

Then the pivot, in roughly increasing order of risk. Build the episode logger and collect a first dataset while driving the hallway by hand, which is useful whichever model wins. Stand up a policy server against a small checkpoint and verify the timing and the safety envelope with a deliberately trivial policy before running a real one. Only then post-train and evaluate. Build the second car from the `hardware/` folder in parallel; it is a parts order and a weekend rather than a research risk, and every agent-to-agent question needs it.

Two warnings for whoever picks this up. Keep the battery connector keyed or marked:

the VESC dies instantly on reversed leads, and a replacement took a week to arrive. And read `hardware/RUNBOOK.md` before touching the VESC's application settings, because the shared PPM/servo pin will look like a dead motor controller to anyone who does not know about it.

Acknowledgments

[To be completed: research teacher, lab, teammates who built the chassis and the simulation stack, and whoever soldered the phase leads at three in the morning.]

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